

Ahmed Fathy

Software Engineer

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EDUCATION

- Cairo University, Faculty of Engineering** Giza, Egypt
B.Sc. in Computer Engineering; GPA: 3.7 2022 – 2027

EXPERIENCE

- Al Nada Scientific Office** | *Software Developer (Part-time)* Cairo, Egypt | June 2022 – 2025
 - ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
 - Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

PROJECTS

- 5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* 2025
 - Core Architecture:** Designed a 32-bit 5-stage pipelined processor in VHDL, with a custom ISA of over 25 instructions.
 - Hazard Resolution:** Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
 - Verification:** Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* 2025
 - Architecture:** Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
 - Memory Hierarchy:** Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
 - Physical Design:** Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* 2023
 - Core Logic:** Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
 - Datapath:** Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
 - Validation:** Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- Tactic (Robotics & AI)** | *Embedded Systems* 2024
 - System Integration:** Developed browser-controlled robotics with AI vision and low-latency control
 - Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
 - Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- Autonomous Wall-Following Robot** | *Embedded Systems & C/C++* 2026
 - System Architecture:** Designed a 5-state FSM to manage autonomous navigation, turning, and data flow.
 - Low-Level Firmware:** Programmed STM32 microcontroller at the register level to optimize hardware timer usage.
 - Control & Integration:** Combined proportional control, ultrasonic/gyro sensors, and WiFi Module for tracking.
- Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* 2024
 - Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
 - Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- Machine Fault Listener System** | *Machine Learning & Audio Processing* 2026
 - Pipeline & Architecture:** Created an end-to-end Machine Learning pipeline to classify industrial audio into six operational states.
 - Optimization:** Optimized a deep learning model for accuracy and fast inference within a 2B-parameter limit.

STUDENT ACTIVITIES

- Physics Helping District (PhD)** | *Team Leader* Cairo | June 2022 – 2024
 - Mentorship:** Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- NASA Space Apps Challenge** | *Participant – Project: "Orrery"* Cairo | Oct 2024
 - Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

TECHNICAL SKILLS

- Digital Design & Verification:** Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design

- **Development Tools & Platforms:** ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages:** C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly